msvtx_nTGCOuterStation

MC vs data · overview · rank 4 / 98

rank 4 / 98 by endcap D_equal

used by: neither NN list

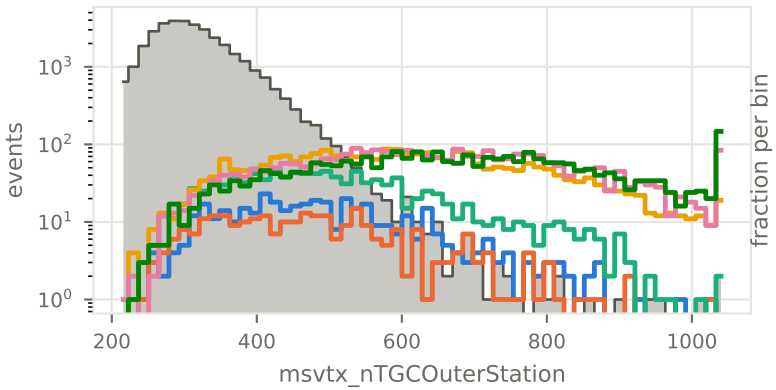
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.725	0.022	0.785	0.666	0.612	0.678	0.785	0.826	0.839	34,970	4/98
barrel	0.063	0.051	0.080	0.053	0.044	0.069	0.063	0.096	0.095	1,089	84/98
all	0.468	0.018	0.505	0.453	0.371	0.449	0.532	0.514	0.529	36,059	11/98

msvtx_nTGCOuterStation

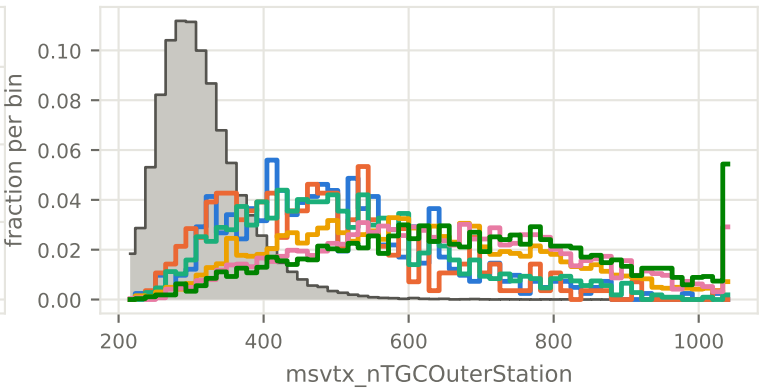
region: ENDCAP · rank 4 / 98

D_equal 0.725 (noise 0.022) D_pooled 0.785 min 0.612 (mH125_mS55) max 0.839 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

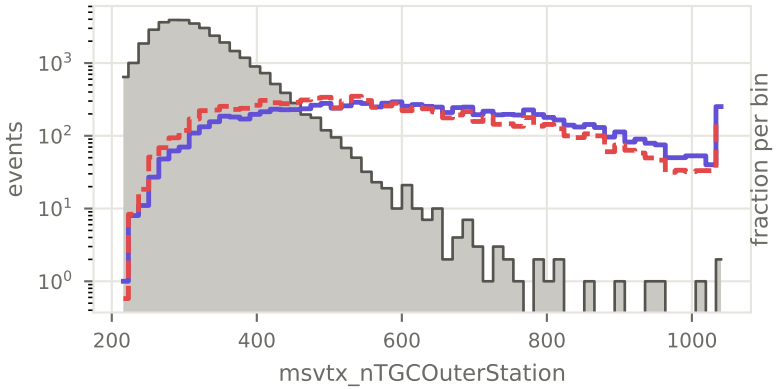


1b · each mass point vs background · normalized

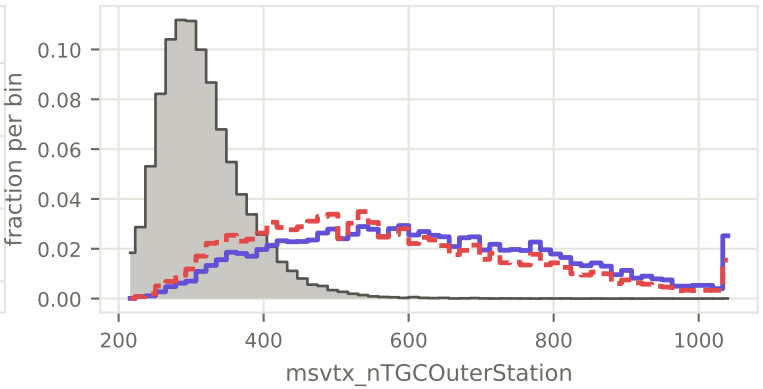


1 · each mass point
background n 34,970
mH125_mS15 D 0.666 n 411
mH125_mS55 D 0.612 n 281
mH200_mS50 D 0.678 n 1,071
mH400_mS100 D 0.785 n 2,648
mH600_mS150 D 0.826 n 2,882
mH1000_mS275 D 0.839 n 2,703

2a · combined signal vs background · raw

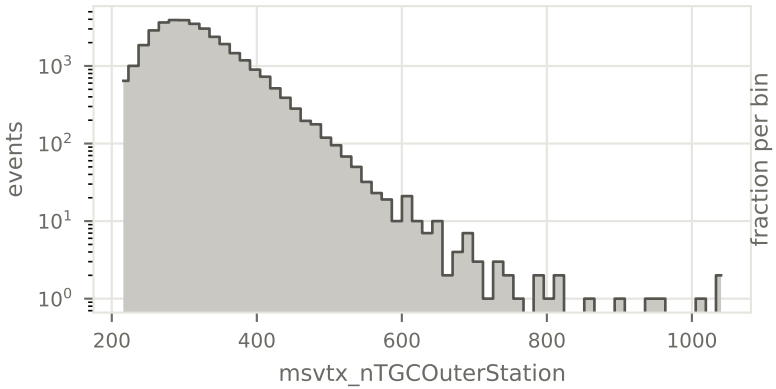


2b · combined signal vs background · normalized

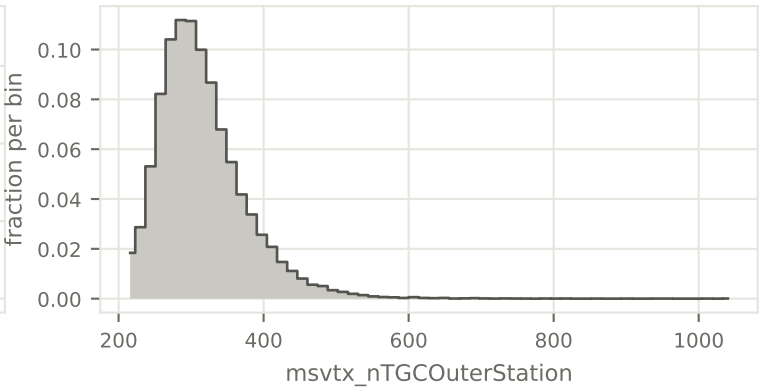


2 · combined signal
background n 34,970
combined, pooled D 0.785 n 9,996
equal weight per mass D 0.725

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

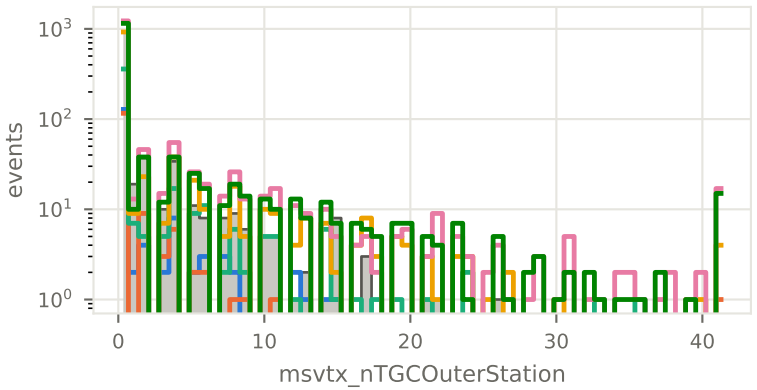
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCOuterStation

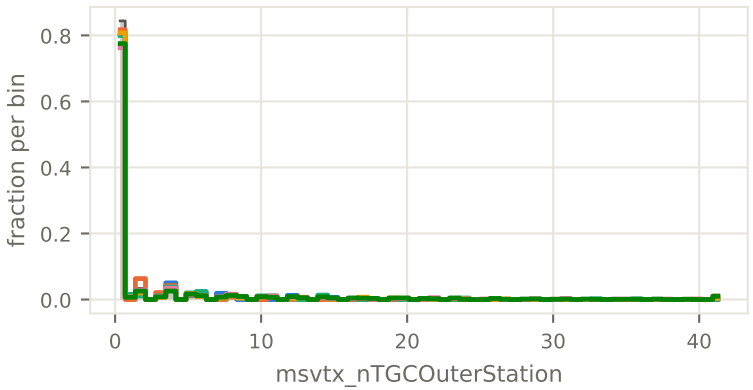
region: BARREL · rank 4 / 98

D_equal 0.063 (noise 0.051) D_pooled 0.080 min 0.044 (mH125_mS55) max 0.096 (mH600_mS150) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

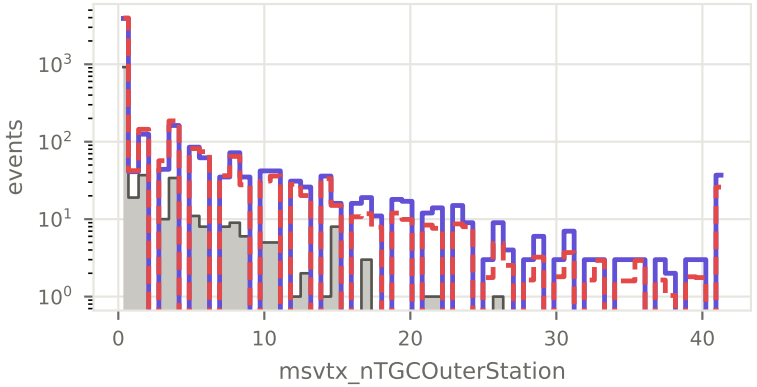


1b · each mass point vs background · normalized

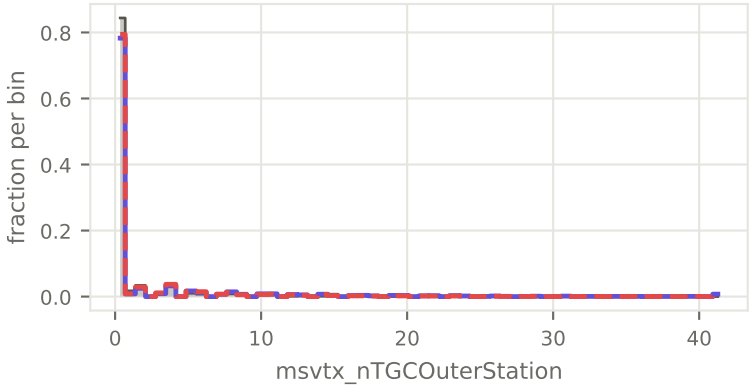


1 · each mass point
background n 1,089
mH125_mS15 D 0.053 n 159
mH125_mS55 D 0.044 n 142
mH200_mS50 D 0.069 n 449
mH400_mS100 D 0.063 n 1,144
mH600_mS150 D 0.096 n 1,608
mH1000_mS275 D 0.095 n 1,485

2a · combined signal vs background · raw

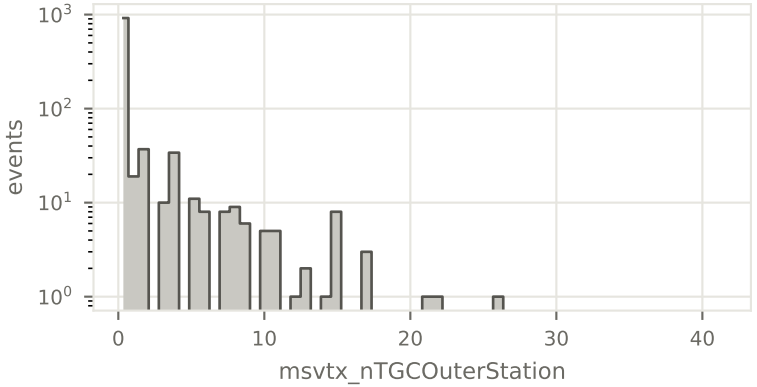


2b · combined signal vs background · normalized

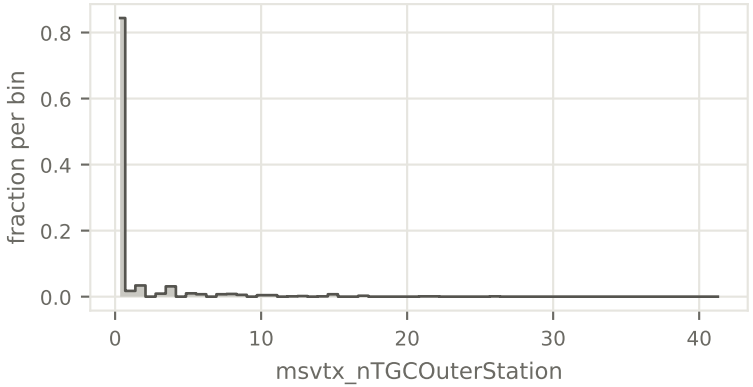


2 · combined signal
background n 1,089
combined, pooled D 0.080 n 4,987
equal weight per mass D 0.063

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

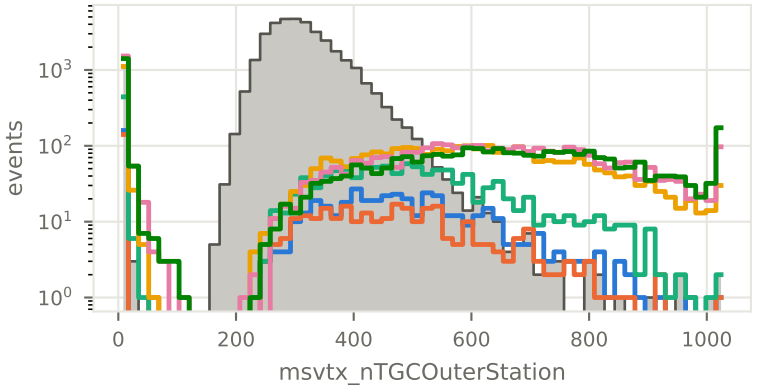
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCOuterStation

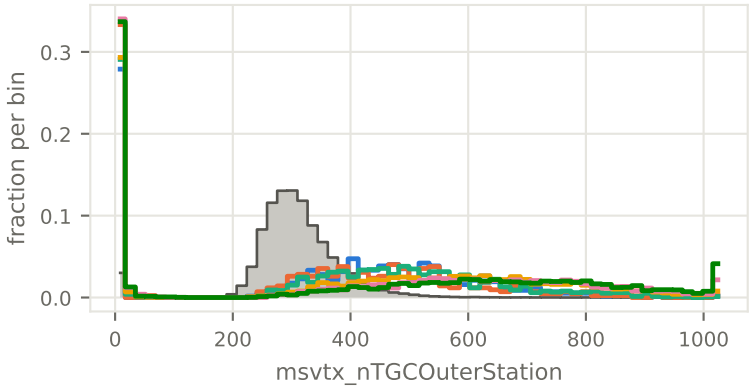
region: ALL · rank 4 / 98

D_equal 0.468 (noise 0.018) D_pooled 0.505 min 0.371 (mH125_mS55) max 0.532 (mH400_mS100) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

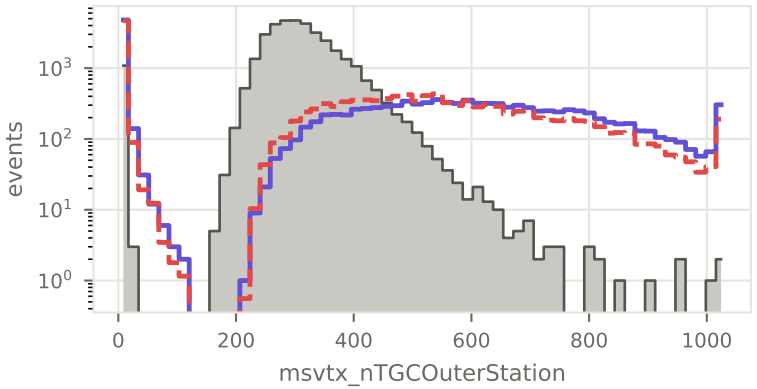


1b · each mass point vs background · normalized

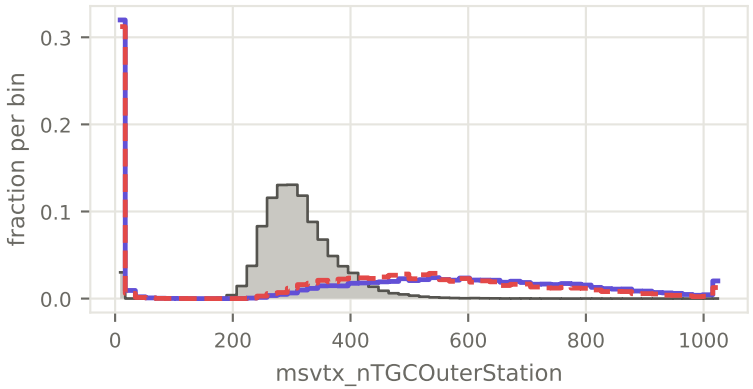


1 · each mass point
background n 36,059
mH125_mS15 D 0.453 n 570
mH125_mS55 D 0.371 n 423
mH200_mS50 D 0.449 n 1,520
mH400_mS100 D 0.532 n 3,792
mH600_mS150 D 0.514 n 4,490
mH1000_mS275 D 0.529 n 4,188

2a · combined signal vs background · raw

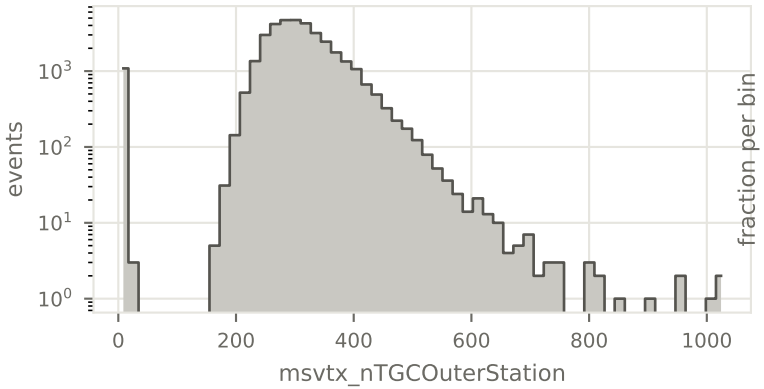


2b · combined signal vs background · normalized

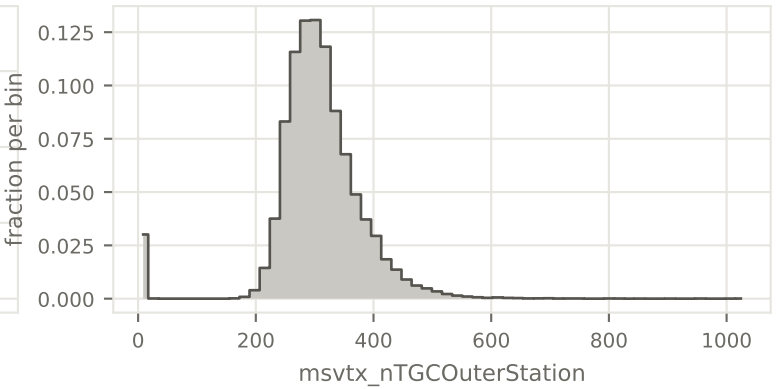


2 · combined signal
background n 36,059
combined, pooled D 0.505 n 14,983
equal weight per mass D 0.468

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed